3.15 X cdf $F(x) = x^2/4$ on $[0,2]$

d. median of X

$F(x) = \frac{1}{2}$ for median

$\frac{x^2}{4} = \frac{1}{2} \Rightarrow x^2 = 2 \quad x = \sqrt{2} \approx 1.4142$

e) pdf of X

$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx}\left(\frac{x^2}{4}\right) = \frac{x}{2}$

so $f(x) = \begin{cases} x/2 & \text{for } 0 \le x \le 2 \\ 0 & \text{for otherwise.} \end{cases}$

f) $E(X) = \int_0^2 x f(x)\,dx = \int_0^2 x\left(\frac{x}{2}\right)dx = \int_0^2 \frac{x^2}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{8}{6} \approx 1.33\ldots$

g) $Var(X) = E(X^2) - E(X)^2$

$\hookrightarrow \int_0^2 x^2 f(x)\,dx = \int_0^2 \frac{x^3}{2}\,dx = \left[\frac{x^4}{8}\right]_0^2 = 2$

$Var(X) = 2 - (\frac{4}{3})^2 = 2 - \frac{16}{9} = \frac{2}{9} \approx 0.22\ldots$

3.21

$M_X(t) = E(e^{tX}) = \sum_{x=1}^{n} e^{tx} f(x)$ $\qquad f(x) = \begin{cases} 1/n & \text{if } x \in \{1,2,\ldots,n\} \\ 0 & \text{otherwise.} \end{cases}$

$= \sum_{x=1}^{n} e^{tx}\left(\frac{1}{n}\right) = \frac{1}{n}\sum_{x=1}^{n} e^{tx} = \frac{1}{n}\left(e^t + (e^t)^2 + (e^t)^3 + \cdots + (e^t)^n\right)$

use geometric sum

$\sum_{x=1}^{n} ar^{x-1} = a\frac{1-r^n}{1-r}$ where $a = e^t$ $r = e^t$

for $t \neq 0$

$M_x(t) = \frac{1}{n} \cdot \frac{e^t(1-e^{tn})}{1-e^t} = \frac{e^t(1-e^{tn})}{n(1-e^t)}$

for $t = 0$

$M_x(t) = \frac{1}{n} \cdot \frac{e^0(1-e^0)}{1-e^0} = \frac{n}{n} = 1$

Final

$M_x(t) = \begin{cases} \frac{e^t(1-e^{nt})}{n(1-e^t)} & t \neq 0 \\ 1 & t = 0 \end{cases}$

3.23 $f(y) = \begin{cases} y & \text{if } y \in [0,1], \\ 2-y & \text{if } y \in [1,2], \\ 0 & \text{otherwise.} \end{cases}$

$\int x e^{tx} dx = \frac{x}{t} e^{tx} - \int \frac{1}{t} e^{tx} dx$

$= \frac{x}{t} e^{tx} - \frac{1}{t^2} e^{tx}$

$= (\frac{x}{t} - \frac{1}{t^2}) e^{tx}$

$M_Y(t) = E(e^{tY}) = \int_{-\infty}^{\infty} e^{ty} f(y)\, dy$

$= \int_0^1 e^{ty} \cdot y \, dy + \int_1^2 e^{ty} \cdot (2-y)\, dy$

$\left(\int e^{tx} \cdot x\, dx = (\frac{x}{t} - \frac{1}{t^2}) e^{tx} \right)$

$= \left[(\frac{y}{t} - \frac{1}{t^2}) e^{ty} \right]_0^1 + 2 \cdot \left[\frac{1}{t} e^{ty} \right]_1^2 - \left[(\frac{y}{t} - \frac{1}{t^2}) e^{ty} \right]_1^2$

$= (\frac{1}{t} - \frac{1}{t^2}) e^t - \{ (0 - \frac{1}{t^2}) \cdot 1 \} + \frac{2}{t}(e^{2t} - e^t) - \left[(\frac{2}{t} - \frac{1}{t^2}) e^{2t} - \{ (\frac{1}{t} - \frac{1}{t^2}) e^t \} \right]$

$= 2(\frac{1}{t} - \frac{1}{t^2}) e^t + \frac{1}{t^2} + \frac{2}{t} e^{2t} - \frac{2}{t} e^t - \frac{2}{t} e^{2t} + \frac{1}{t^2} e^{2t}$

$= -\frac{2}{t^2} e^t + \frac{1}{t^2} + \frac{1}{t^2} e^{2t}$

$= \frac{e^{2t} - 2e^t + 1}{t^2} = \left(\frac{e^t - 1}{t} \right)^2$ if $y \in [0,2]$

3.31 $M_X(t) = \frac{e^{2t}}{1-t^2}$, $M_X'(t) = 2e^{2t}(1-t^2)^{-1} + e^{2t} \cdot 2t(1-t^2)^{-2}$

$= \frac{2e^{2t}}{1-t^2} + \frac{2te^{2t}}{(1-t^2)^2}$

$E(X) = M_X'(0)$

$\rightarrow M_X'(0) = \frac{2e^0}{1} + \frac{0}{1} = \boxed{2}$

$E[X^2] = M_X''(0)$, $M_X''(t) = \frac{d}{dt} M_X'(t) = \frac{4e^{2t}}{1-t^2} + \frac{8te^{2t}}{(1-t^2)^2} + \frac{2e^{2t}}{(1-t^2)^2} + \frac{8t^2 e^{2t}}{(1-t^2)^3}$

$M_X''(0) = 4 + 0 + 2 + 0 = \boxed{6}$

$Var(X) = E(X^2) - E(X)^2 = 6 - 2^2 = \boxed{2}$

3.33

$$M_x(t) = \left[\frac{3}{3-t}\right]^2 = 9(3-t)^{-2}$$

$$E(X) = M'_x(0) = 9(3-t)^{-3}\cdot(-2)\cdot(-1)$$
$$= 18(3-t)^{-3}$$

at $t=0$, $E(X) = \frac{18}{27} = \frac{2}{3}$

$$E(X^2) = M''_x(0) = 18\cdot(-3)(3-t)^{-4}\cdot(-1)$$
$$= 54(3-t)^{-4}$$

at $t=0$, $E(X) = \frac{54}{81} = \frac{2}{3}$

$$Var(X) = E(X^2) - [E(X)]^2$$
$$= \frac{2}{3} - \left(\frac{2}{3}\right)^2$$
$$= \frac{2}{3} - \frac{4}{9}$$
$$Var(X) = \frac{2}{9}$$

3.34 $X \sim Geom(\pi)$

(a) $M_X(t) = E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} P(X=x)$

$$= \sum_{x=0}^{\infty} e^{tx}\,\pi(1-\pi)^x$$
$$= \pi \sum_{x=0}^{\infty} e^{tx}(1-\pi)^x$$

let $r^x = (e^t(1-\pi))^x$, so $r = e^t(1-\pi)$

therefore $= \pi \cdot \frac{1}{1-(1-\pi)e^t}$ for $0 < \pi < 1$

(c) $E(X)$; $M'_X(t) = \frac{\pi(1-\pi)e^t}{[1-(1-\pi)e^t]^2}$, $M'_X(0) = E(X) = \frac{\pi(1-\pi)e^0}{[1-(1-\pi)e^0]^2} = \frac{\pi(1-\pi)}{\pi^2} = \boxed{\frac{1-\pi}{\pi}}$

$E[X^2]$; $M''_X(t) = \frac{d}{dt}M'_X(t) = \frac{\pi(1-\pi)e^t}{[1-(1-\pi)e^t]^2} + \frac{2\pi(1-\pi)^2e^{2t}}{[1-(1-\pi)e^t]^3}$

while $t=0$, $M''_X(0) = E(X^2) = \frac{\pi(1-\pi)}{\pi^2} + \frac{2\pi(1-\pi)^2}{\pi^3}$

$$= \frac{1-\pi}{\pi} + \frac{2(1-\pi)^2}{\pi^2}$$

$$Var(X) = E(X^2) - E(X)^2 = \left(\frac{1-\pi}{\pi} + \frac{2(1-\pi)^2}{\pi^2}\right) - \left(\frac{1-\pi}{\pi}\right)^2$$
$$= \frac{(1-\pi)[2-\pi] - (1-\pi)^2}{\pi^2} = \boxed{\frac{1-\pi}{\pi^2}}$$

3.37

(a) $M_W(t) = \left(\frac{e^t + 1}{2}\right)^{10} = \left(\frac{e^t}{2} + \frac{1}{2}\right)^{10}$

$\Rightarrow X \sim \text{Binom}(10, 0.5)$

(b) $M_X(t) = e^{t + t^2/2} \Rightarrow \mu = 1$ and $\sigma^2 = 1$

$\Rightarrow X \sim \text{Norm}(1, 1)$

(c) $M_Y(t) = \left(\frac{1}{1-2t}\right) \Rightarrow \frac{1}{\lambda} = 2 \Rightarrow \lambda = \frac{1}{2}$

$\Rightarrow X \sim \text{Exp}(1/2)$

(d) $M_Z(t) = \left[\frac{1}{1-2t}\right]^3 \Rightarrow \alpha = 3$ and $\beta = 2$

$\Rightarrow X \sim \text{Gamma}(3, 2)$

3.38) Suppose $X \sim \text{Gamma}(\alpha, \lambda)$ Does $3X$ have a gamma distribution? What is its parameters?

$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^{\alpha}$ $M_{3X}(t) = M_Y(3t) = \left(\frac{\lambda}{\lambda - 3t}\right)^{\alpha} = \left(\frac{\frac{\lambda}{3}}{\frac{\lambda}{3} - t}\right)^{\alpha}$

Yes $3X$ is gamma,

$3X \sim \text{Gamma}\left(\alpha, \frac{\lambda}{3}\right)$

3.39

(a) $\text{Exp}(\lambda = 1)$

$\mu = \frac{1}{\lambda} = 1$

$\sigma = \frac{1}{\lambda} = 1$

$[0, 2]$

cdf $F(x) = 1 - e^{-x}$

$F(2) - F(0) = 1 - e^{-2} - 1 + 1$

$\approx 14.5\%$

(b) $\mu = \frac{1}{2}$, $\sigma = \frac{1}{2}$

interval $(0, 1)$

$F(x) = 1 - e^{-2x}$

$= 1 - e^{-2x}$

$= 86.5\%$

(c) Unif $(0, 1)$

$\mu = \frac{1}{2}$ $\sigma = \sqrt{\frac{1}{12}}$

$p = 2\sigma = \frac{1}{\sqrt{3}} = 57.7\%$

(d) $\mu = \frac{\alpha}{\alpha+\beta} = \frac{2}{6} = \frac{1}{3}$

$\sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{2}{63}$

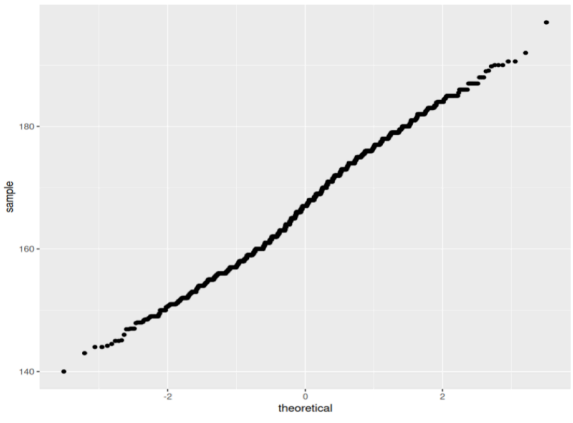
~~$f(x) = 20x(1-x)^3$~~

R pbeta(0.8418, 2, 4) − pbeta(0.1582, 2, 4)

$= 65.2\%$.

3.46

(a) The graph is the normal-quantile plot of the heights of all subjects in data Pheno. The points do not form a perfect straight line, but they follow a roughly linear pattern, suggesting that the heights data are approximately normally distributed.



(b) Both QQ plots show an approximately linear pattern. This suggests that the height distributions for both sexes are approximately normal.

